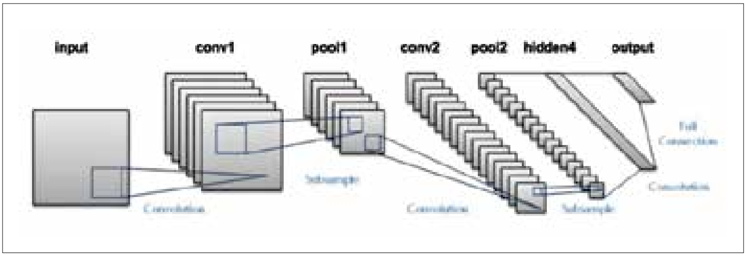

Structure of the LeNet-5.

LeNet-5

LeNet 5 was created in 1998, therefore, it is one of the pioneering CNN systems. It hosts a seven-level convolutional network that specifically classifies digits. This CNN was adopted by a host of banks and banking services in order to be able to recognise handwritten numbers on cheques that were digitised in 32x32 pixel monochrome input images. If one wants to process images with higher resolution, larger and more layers would be required. Hence, we see that this technology was constrained due to limited computing resources.

AlexNet

This CNN was incepted in 2012 and it successfully outperformed all the previous competitors. It did so by coming on top of the challenge by reducing top-5 error from 26% to approximately 15%. The network boasted a similar architecture to LeNet but its championing factor was that it was significantly deeper, with added filters per layer along with stacked convolutional layers.

ZFNet

ZFNet is a popular CNN that was crowned the ILSVRC (CNN competition) 2013 winner. It managed to achieve a top-5 error rate of only 14.8%. This achievement of ZFNet was accomplished by tweaking the hyper-parameters of AlexNet while retaining the same structure. Additional deep learning elements were also added to further enhance the effectiveness and accuracy of ZFNet.

GoogleNet/Inception

GoogleNet or Inception by Google was the winner of the ILSVRC in 2014. Astoundingly, it managed to achieve a top-5 error rate of a mere 6.67%!